

Chapter-2

*Optical fiber communication system:

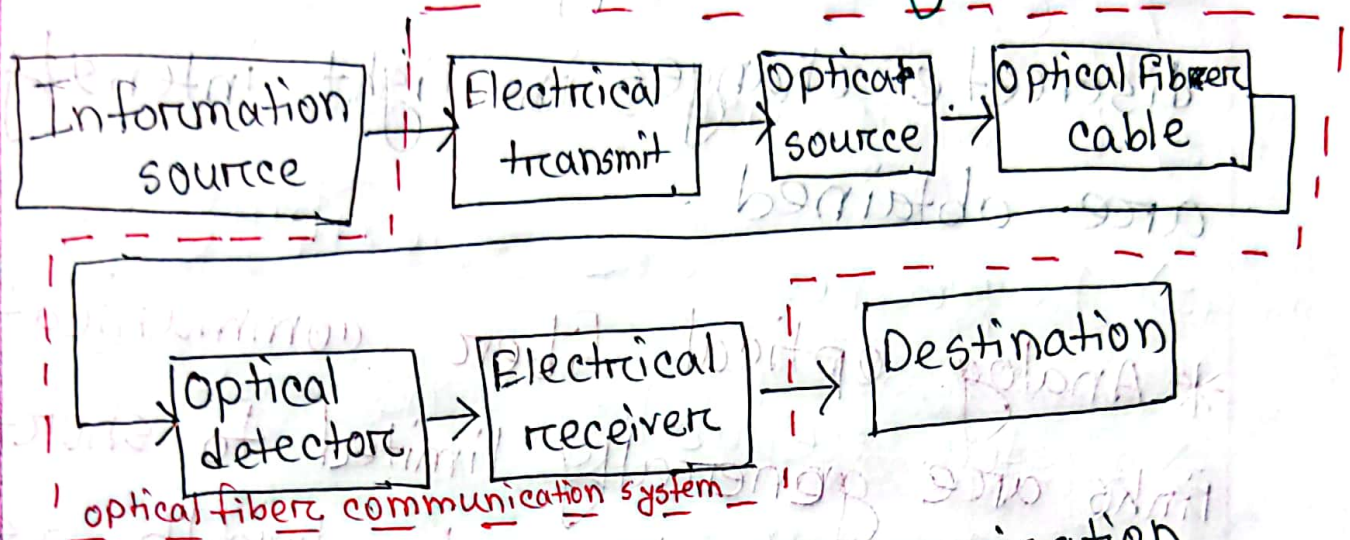


Fig: Optical Fiber Communication

Here, information source provides an electrical signal to a transmitter comprising an electrical stage which drives an optical source to give modulation of lightwave carrier. Optical source may be either semiconductor laser or light emitting diode (LED). The optical carrier may be modulated using either analog or digital information signal. In analog modulation involves the variation of the light emitted from optical source in a continuous

manner and in digital modulation, discrete changes in light intensity are obtained.

* Analog optical fiber communication links are generally limited to shorter distance and lower bandwidth as it is less efficient and require far higher signal to noise ratio at the receiver.

Carrier wave + signal wave - modulation

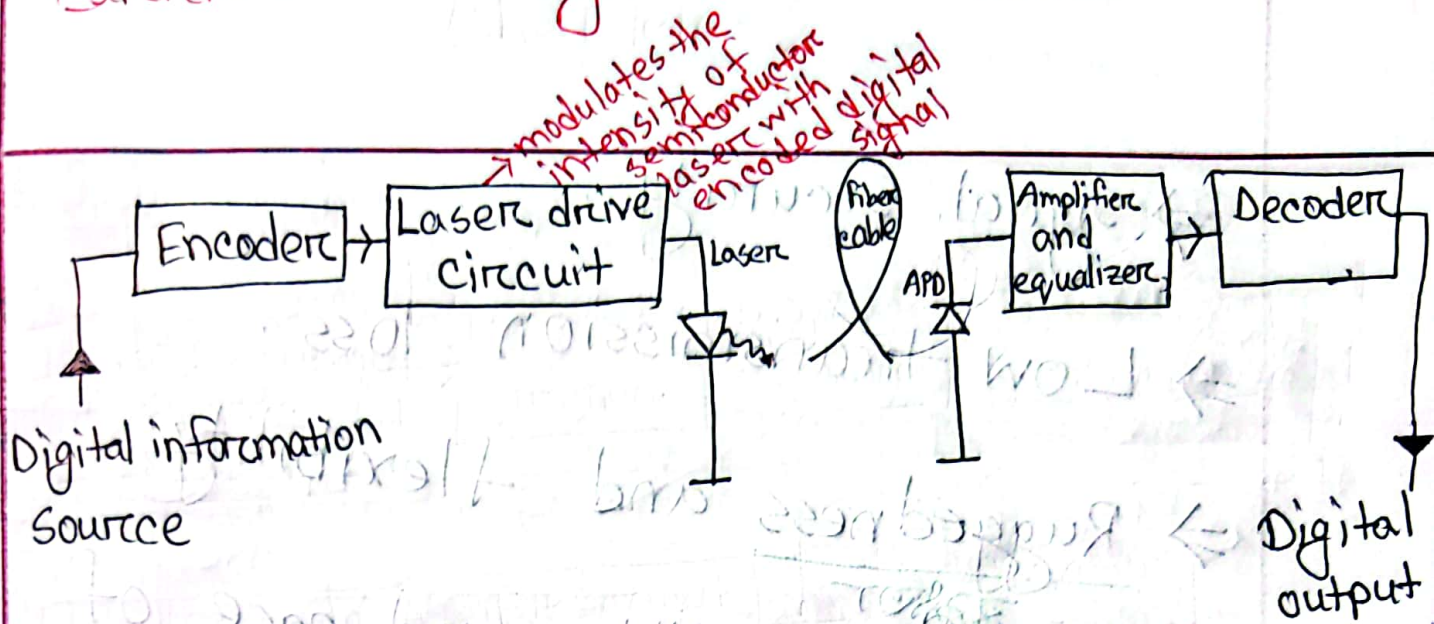


Fig: A digital optical fiber link using a semiconductor laser source and an avalanche photodiode (APD) Diode detector.

* Advantage of optical fiber communication:

⇒ Enormous potential bandwidth.
Optical carrier frequency in the range 10^{13} to 10^{16} Hz.

⇒ Small size and weight.

⇒ Electrical isolation

Optical fibers are fabricated from glass or plastic polymer. They are electrical insulators.

⇒ Immunity to interference and crosstalk.
Free from electromagnetic interference and radio-frequency interference.

electrically
noisy
environment
avoid

⇒ Signal security

⇒ Low transmission loss

⇒ Ruggedness and flexibility

⇒ System reliability and ease of maintenance

⇒ Potential low cost

⇒ Error-free transmission

⇒ Small size and weight

⇒ Electrical isolation

⇒ Immunity to electromagnetic interference

⇒ Low power consumption

⇒ High bandwidth

Chapter - 2

Optical fiber waveguide

The refractive index of a medium is defined as the ratio of the velocity of light in a vacuum to the velocity of light to the medium.

* Total internal reflection:

The angles of incidence ϕ_1 and refraction ϕ_2 are related to each other and to the refractive indices of the dielectrics by Snell's law of refraction which states that:

$$\rightarrow n_1 \sin \phi_1 = n_2 \sin \phi_2$$

$$\text{or, } \frac{\sin \phi_1}{\sin \phi_2} = \frac{n_2}{n_1}$$

* Types of optical fibers:

1. step index optical fibers
2. Graded index optical fibers

Snell's law: The angle of incidence ϕ_1 and refraction angle ϕ_2 are related to each other and to the refractive indices of the dielectrics by Snell's law of refraction.

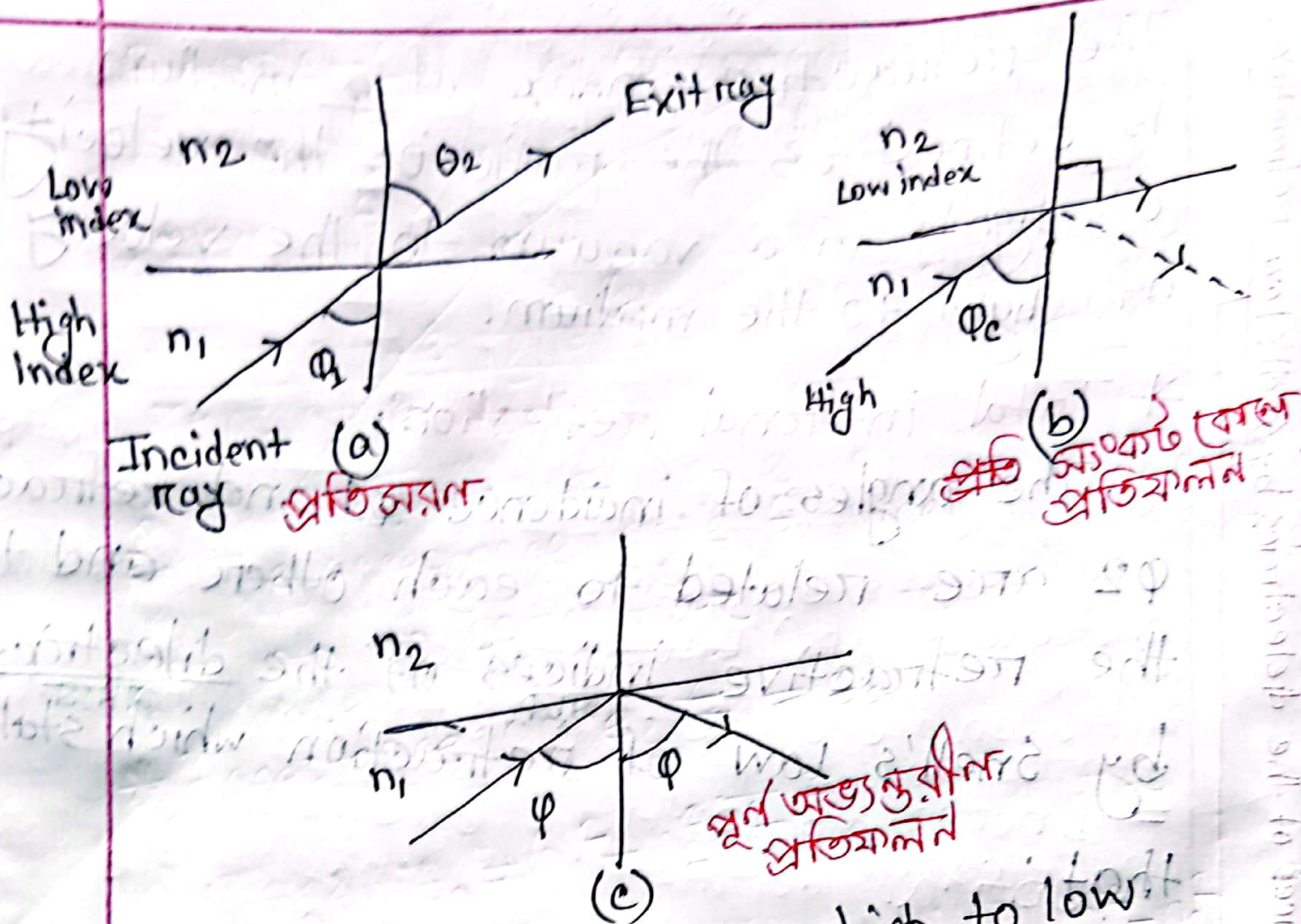


Fig: Light ray incident on a high to low refractive index (a) refraction (b) the limiting case of refraction showing the critical ray at an angle ϕ_c (c) total internal reflection where $\phi > \phi_c$.

$$\sin \phi_c = \frac{n_2}{n_1}$$

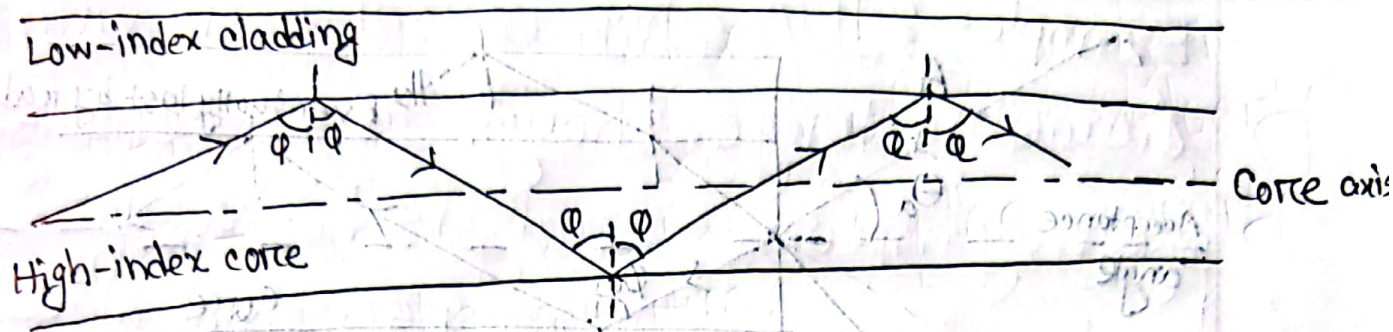


Fig: The transmission of a light ray in a perfect optical fiber

* At angles of incidence greater than the critical angle the light is reflected back into the originating dielectric medium with high efficiency.

* Acceptance angle:

The maximum angle to the axis at which light may enter the fiber in order to be propagated is called acceptance angle.

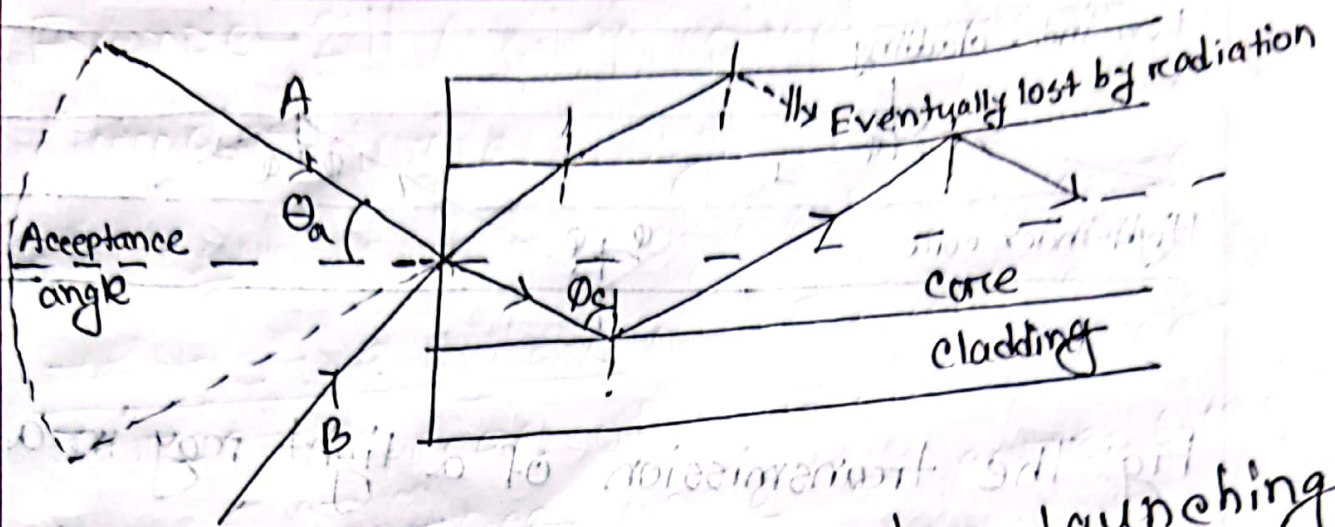


Fig. The acceptance angle θ_a when launching light into an optical fiber.

* Acceptance angle, $\theta_a = \theta_c$

$$n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

* Numerical aperture:

$$NA = n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

$$NA = n_1 (2\Delta)^{\frac{1}{2}}$$

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\approx \frac{n_1 - n_2}{n_1}$$

$$\Delta ABC \rightarrow \theta_1 = \frac{\pi}{2} \text{ --- (1)}$$

$$(1) \Rightarrow n_0 \sin \theta_a = n_1 \sin \left(\frac{\pi}{2} - \phi \right)$$

$$\Rightarrow n_0 \sin \theta_a = n_1 \cos \phi$$

$$\Rightarrow n_0 \sin \theta_a = n_1 (1 - \sin^2 \phi)^{\frac{1}{2}}$$

$$\Rightarrow n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

Because, when the limiting cone for total internal reflection is considered, ϕ because equal to the ~~critical~~ critical angle for the core-cladding interface.

$$\Rightarrow NA = n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

$$NA = (2n_1^2 \Delta)^{\frac{1}{2}}$$

$$= n_1 (2\Delta)^{\frac{1}{2}}$$

$$\because \Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\frac{1}{2} \left(\frac{c}{v_1} - \frac{c}{v_2} \right) = \Delta$$

$$0.30 =$$

Example-2] A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and cladding refractive index of 1.47. Determine

- the ~~acceptance~~^{critical} angle at core-cladding
- NA for fiber
- acceptance angle in air for fiber

Solution:

- The critical angle ϕ_c at core-cladding

$$\phi_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1.47}{1.50}$$

$$= 78.5^\circ$$

- The NA is:

$$NA = (n_1^2 - n_2^2)^{\frac{1}{2}}$$

$$= \left\{ (1.50)^2 - (1.47)^2 \right\}^{\frac{1}{2}}$$

$$= 0.30$$

c) acceptance angle:

$$\theta_a = \sin^{-1} n_0 \sin^{-1} \left[\frac{(n_1^2 - n_2^2)^{1/2}}{n_0} \right]$$

$$\theta_a = \frac{\sin^{-1} (n_1^2 - n_2^2)^{1/2}}{n_0}$$

$$= \sin^{-1} NA \quad \left[\begin{array}{l} NA = (n_1^2 - n_2^2)^{1/2} \\ n_0 = 1 \end{array} \right]$$

$$= \sin^{-1} (0.30)$$

$$= 17.458^\circ$$

Example: 2.2] A typical refractive index difference for an optical fiber designed for long distance transmission is 1%. Estimate the NA and solid acceptance angle in air for fiber when core index is 1.46. Calculate critical angle at core-cladding.

$$1 - 1 = \frac{1.46}{1.47}$$

$$1.01 - 1 =$$

$$0.01 =$$

Solution: difference

$$\text{Hence, } \Delta = 1\% = 0.01$$

$$NA = n_1 (2\Delta)^{\frac{1}{2}} \\ = 1.46 (2 \times 0.01)^{\frac{1}{2}} \\ = 0.21$$

For small angles, the solid angle in air is:

$$\Sigma \approx \pi \theta_a^2 = \pi \sin^2 \theta_a$$

$$\Sigma \approx \pi (NA)^2 = \pi \times (0.21)^2 \\ = 0.13 \text{ rad}$$

for relative refractive index difference

$$\Delta \approx \frac{n_1 - n_2}{n_1} = 1 - \frac{n_2}{n_1}$$

Hence

$$\frac{n_2}{n_1} = 1 - \Delta \\ = 1 - 0.01 \\ = 0.99$$

critical angle;

$$\phi_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} 0.99 = 81.9^\circ$$

* Skew ray

A skew ray refers to a type of light that travels through the fiber core in a helical or spiral path.

Example

* The acceptance angle for skew ray,

$$n_0 \sin \theta_{as} \cos \gamma = (n_1^2 - n_2^2)^{\frac{1}{2}} = NA$$

in case of fiber in air ($n_0 = 1$):

$$\sin \theta_{as} \cos \gamma = NA$$

$$NA = \gamma \cos \theta_{as}$$

$$\frac{NA}{\cos \theta_{as}} = \gamma$$

$$\left(\frac{NA}{\cos \theta_{as}} \right)^2 = \gamma^2$$

Example 12.3] An optical fiber in air has an NA of 0.4. Compare acceptance angle for meridional rays with that for skew ray which change direction by 100° at each reflection.

Solution:

acceptance angle for meridional ray:

$$n_0 \sin \theta_a = NA$$

$$\Rightarrow \theta_a = \sin^{-1}(NA)$$

$$NA = \sin^{-1}(0.4)$$

$$23.6^\circ$$

The skew ray change direction by 100° at each reflection, therefore, $\gamma = 50^\circ$

acceptance angle for skew ray:

$$n_0 \sin \theta_{as} \cos \gamma = NA$$

$$\Rightarrow \sin \theta_{as} = \frac{NA}{\cos \gamma}$$

$$\Rightarrow \theta_{as} = \sin^{-1}\left(\frac{NA}{\cos \gamma}\right)$$

$$= \sin^{-1} \left(\frac{0.4}{\cos 50^\circ} \right)$$

$$= 38.5^\circ$$

$$\therefore \frac{\theta_{as}}{\theta_a} = \frac{38.5}{23.5}$$

$$\therefore \theta_{as} - \theta_a = 38.5 - 23.5 = 15^\circ$$

Here, acceptance angle for skew ray is 15° greater than meridional ray. However it must be noted that we have only compared acceptance angle of one particular skew ray path. When light input to fiber is at angle to the fiber axis, it is possible that γ will vary from ~~zero~~ 0 to 90° for meridional rays which enter at core-cladding interface giving acceptance of skew rays over a conical half angle of $\frac{\pi}{2}$ radians.

Chapter - 3

* Attenuation:

It refers to the reduction in power of the light signal as it travels through the fiber. It also called transmission losses. It determines maximum transmission distance.

input optical power = P_i

output " " = P_o

Number of decibels (dB) = $10 \log_{10} \frac{P_i}{P_o}$

powers and roots reduce to multiplication and division.

In addition and subtraction require a conversion to numerical values which may be obtained:

numerical value, $\frac{P_i}{P_o} = 10^{(dB/10)}$

$$\alpha_{dB} L = 10 \log_{10} \frac{P_i}{P_o} \text{ dB km}^{-1}$$

α_{dB} is the signal attenuation per unit length in decibels.

Example: 3.11 When the mean optical power launched into an 8 km length of fiber is 120 μ W the mean optical power at the fiber output is 3 μ W. Determine:

a) the overall signal attenuation in decibels through the fiber assuming there are no connectors or splices.

b) the signal attenuation per kilometer

c) overall signal attenuation for 10 km optical link using the same fiber with splices at 1 km intervals, each giving attenuation of 1 dB

d) Numerical input/output power ratio in (c)

Solution:

a) The overall signal attenuation:

$$= 10 \log_{10} \frac{P_i}{P_o}$$

$$= 10 \log_{10} \frac{120 \times 10^{-6}}{3 \times 10^{-6}}$$

$$= 10 \log_{10} 40$$

$$= 16.0 \text{ dB}$$

b) The signal attenuation per kilometre:

$$\alpha_{\text{dB}} L = 10 \log_{10} \frac{P_i}{P_o}$$

$$\alpha_{\text{dB}} = \frac{16.0}{L}$$

$$= \frac{16.0}{8}$$

$$= 2.0 \text{ dB km}^{-1}$$

c) as $\alpha_{dB} = 2 \text{ dB km}^{-1}$, the loss incurred along 10 km of the fiber is

$$\alpha_{dB} \times L = 2 \times 10 = 20 \text{ dB}$$

However, the link also has nine splices each with an attenuation of 1 dB. Therefore, the loss due to the splices is 9 dB
Overall signal attenuation $\epsilon = 20 + 9$

$$= 29 \text{ dB}$$

d) Numerical value for input/output power

ratio

$$\frac{P_i}{P_o} = 10^{\left(\frac{\text{dB}}{10}\right)}$$

$$= 10^{\left(\frac{29}{10}\right)}$$

$$= 794.3$$

* A number of mechanism are responsible for signal attenuation. They are:

⇒ The material composition

⇒ purification technique.

⇒ The preparation.

⇒ The waveguide structure.

These mechanism may be categorized within several major areas:

⇒ Material absorption.

⇒ material scattering.

⇒ Curve and microbending loss.

⇒ mode coupling radiation loss.

⇒ Losses due to leaky modes.

* Material Absorption:

It is a loss mechanism related to the material composition and fabrication process for fiber, which results in dissipation of some of the transmitted optical power as heat in the waveguide. Absorption are two types:

1. Intrinsic Absorption: Caused by the interaction with one or more of the major components of glass.

2. Extrinsic Absorption: Caused by impurities within the glass. Impurities are Cr^{3+} , Cu^{2+} , OH^- ion.

2. Mie Scattering : caused by large particle comparable in size to the wavelength

* Linear scattering losses :

Linear scattering losses cause the transfer of some or all of the optical power contained within one propagating mode to be transferred linearly into a different mode. It is two types.

1. Rayleigh scattering:

Rayleigh is the elastic scattering of light or other electromagnetic radiation by particles much smaller than wavelength of the radiation.

$$\gamma_R = \frac{8\pi^3}{3\lambda^4} n^2 p^2 \beta_c K T_F$$

γ_R = Rayleigh scattering coefficient
 λ = optical wavelength
 n = Refractive index of medium.

p = average photoelastic coefficient

β_c = isothermal compressibility at a fictive temperature T_f

K = Boltzmann's constant.

Rayleigh scattering coefficient is related to transmission loss factor of the fiber

$$L = \exp(-\alpha_R L)$$

L = length of the fiber

*Example 3.2 Silica has an estimated fictive temperature of 1400K with an isothermal compressibility of $7 \times 10^{-11} \text{ m}^2 \text{ N}^{-1}$. The refractive index and photoelastic coefficient for silica are 1.46 and 0.286 . Determine theoretical attenuation in decibels per kilometer

due to fundamental Rayleigh scattering in silica at optical wavelengths of 0.63 and 1.00 and 1.30 μm . Boltzmann's constant is $\frac{1.381 \times 10^{-21} \text{ J K}^{-1}}{\text{K}}$

Solution:

$$\gamma_R = \frac{8 \pi^3 n^8 p^2 \beta_c K T_F}{3 \lambda^4}$$

$$= \frac{8 \pi^3 \times (1.46)^8 \times (0.286)^2 \times 7 \times 10^{-11} \times 1.381 \times 10^{-21} \times 1400}{3 \times \lambda^4}$$

$$= \frac{1.895 \times 10^{-28}}{\lambda^4} \text{ m}^{-1}$$

at wavelength of 0.63 μm :

$$\gamma_R = \frac{1.895 \times 10^{-28}}{(0.63 \times 10^{-6})^4}$$

$$= 1.199 \times 10^{-3} \text{ m}^{-1}$$

transmission loss factor for 1 kilometer

$$L_{km} = \exp(-\gamma_R L) = \exp(-1.199 \times 10^{-3} \times 10^3)$$

$$L_{km} = \exp(-1.199) = 0.301$$

The attenuation due to Rayleigh scattering

$$= 10 \log_{10} \left(\frac{1}{L_{km}} \right)$$

$$= 52 \text{ dB km}^{-1}$$

At wavelength of 1.0 μm :

$$\gamma_R = \frac{1.895 \times 10^{-28}}{10^{-24}}$$

$$= 1.895 \times 10^{-4} \text{ m}^{-1}$$

$$L_{km} = \exp(-1.895 \times 10^{-4} \times 10^3)$$

$$= 0.827$$

$$\therefore \text{Attenuation} = 10 \log_{10} \left(\frac{1}{L_{km}} \right)$$

$$= 0.8 \text{ dB km}^{-1}$$

at wavelength $1.30 \mu\text{m}$

$$\gamma_R = \frac{1.895 \times 10^{-28}}{(1.30 \times 10^6)^4} = 0.664 \times 10^{-4}$$

$$\rightarrow I_{\text{km}} = \exp(-\gamma_R L) = \exp(-0.664 \times 10^{-4} \times 10^3)$$

$$= 10.936$$

$$\therefore \text{Attenuation} = 10 \log_{10} \left(\frac{1}{I_{\text{km}}} \right)$$

$$= 0.3 \text{ dB km}^{-1}$$

*** Fiber bend loss:**

Optical fibers suffer radiation losses at bends or curves on their paths. This is due to the energy in the evanescent field at the bend exceeding the velocity of light in the cladding and hence the guidance mechanism is inhibited which causes light energy to be

radiated from fiber.

Radiation attenuation coefficient:

$$\alpha_r = c_1 \exp(-c_2 R)$$

R = radius of curvature of fiber bend

c_1, c_2 = constant

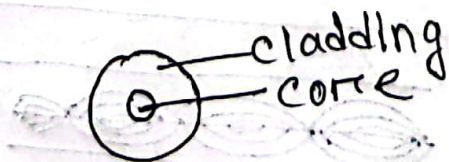
in multimode fibers at a critical radius of curvature R_c

$$R_c \approx \frac{3n_1^2 \lambda}{4\pi(n_1^2 - n_2^2)}$$

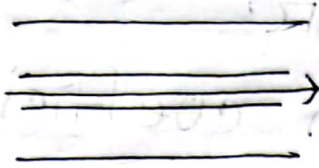
* Different mode fibers:

1. Single mode: It has a small diameter core that allows one mode of light to propagate. It is used in long distance.

distance



Light path

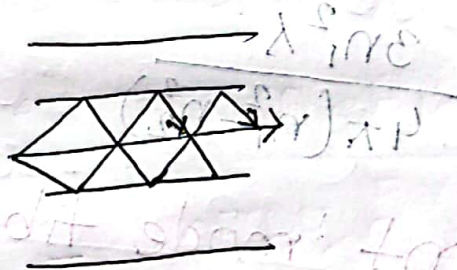


2. Multimode-step index! It has a large diametrical core that allows multiple modes of light to propagate.



core

Light path

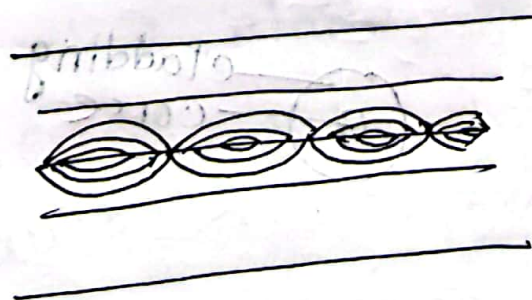


3. Multimode-Graded index! It has a large diametrical core that allows multiple modes of light to propagate.



core

Light path



→ विकृति

* Dispersion:

Dispersion refers to the spreading of light pulse as they travel down the fiber. It causes distortion for both digital and analog transmission.

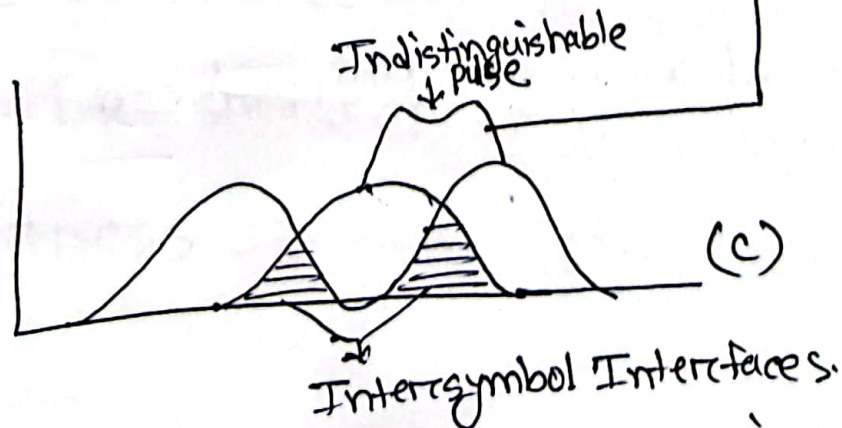
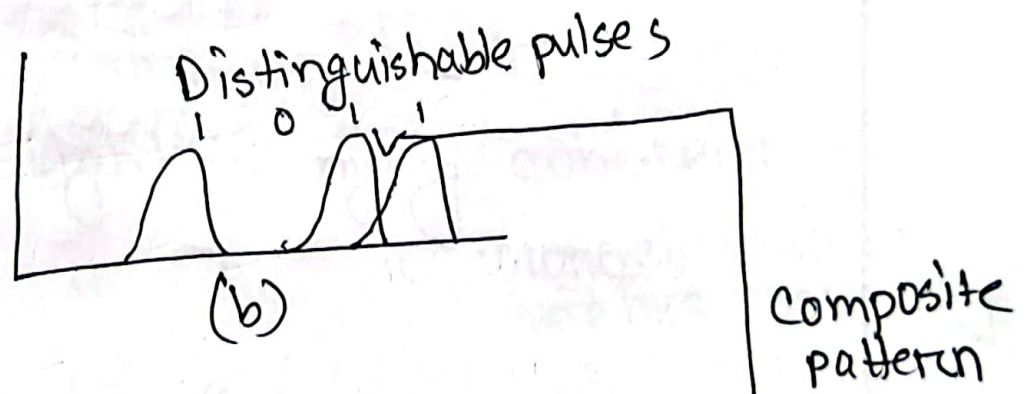
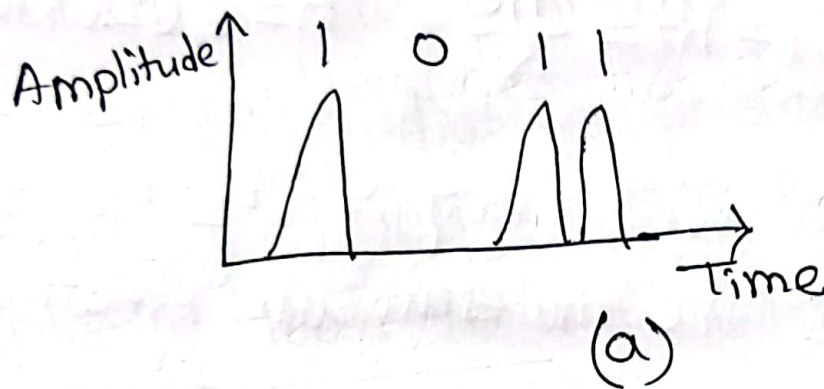
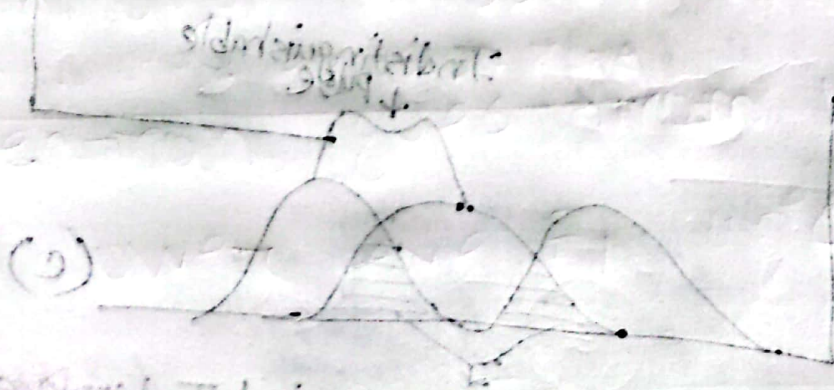
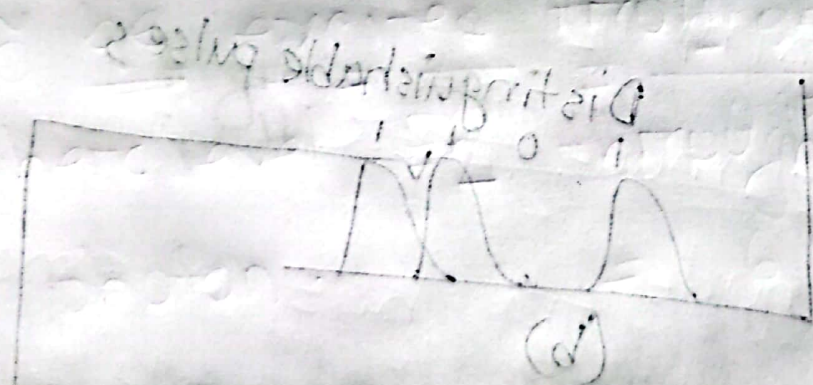
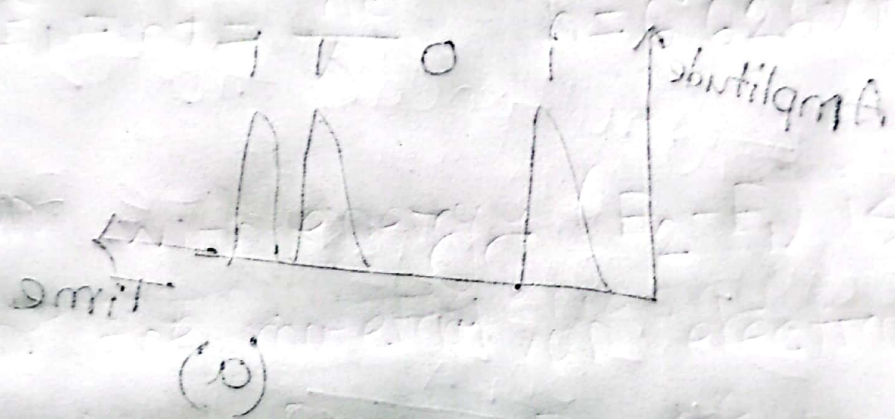


Fig: An illustration using digital bit 1011 of broadening of light

pulse. a) fiber input b) fiber output at a
 distance L_1 , c) fiber output at distance
 $L_2 > L_1$



substituting for the critical radius of curvature for the single mode fiber gives

$$R_c \approx \frac{20\lambda}{(n_1 - n_2)^{3/2}} \times \frac{1}{2} = \frac{20 \times 1.55 \times 10^{-6}}{(1.500 - 1.48)^{3/2}} = 3.4 \text{ mm}$$

pulse. a) fiber input b) fiber output at a distance L , c) fiber output at distance $L_2 > L_1$

Ex. 3.4 Two step index fibers exhibit following parameters:
 a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm .

b) a 8 μm core diameter single mode fiber with a core refractive index the same as (a) relative refractive index difference of 0.3% and an operating wavelength of 1.55 μm .
 Estimate the critical radius of curvature at which large bend losses occur in both cases.

Soln: a) The relative refractive index difference:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

Hence, $n_2^2 = n_1^2 - 2\Delta n_1^2 = (1.500)^2 - 2 \times 0.03 \times (1.500)^2 = 2.25 - 0.06 \times 2.25 = 2.115$

Multimode fiber critical radius of curvature:

$$R_c \approx \frac{3n_1^2 \lambda}{4\pi(n_1^2 - n_2^2)^{3/2}} = \frac{3 \times 2.25 \times 0.82 \times 10^{-6}}{4\pi(2.25 - 2.115)^{3/2}} = 9 \mu\text{m}$$

b) Again, $\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \Rightarrow n_2^2 = n_1^2 - 2\Delta n_1^2 = (1.500)^2 - 2 \times 0.003 \times (1.500)^2 = 2.237$

The cutoff wavelength for single mode fiber:

$$\lambda_c = \frac{2\pi a n_1 (2\Delta)^{1/2}}{2.405} = \frac{2\pi \times 4 \times 10^{-6} \times 1.500 \times (2 \times 0.003)^{1/2}}{2.405} = 1.214 \mu\text{m}$$

Chapter 14

* The energy difference due to the transition of electron leads two process:

1. Radiative process: In this process

photons are emitted.

2. Non radiative process:

This process

typically lead to the creating of heat.

* characteristic of laser:

1. Laser light is highly coherent

2. " " " " Monochromatic
একই দিকে একটা রশ্মি
যায়.

3. can be sharply focused

4. Brightness is high

* absorption and emission of light:

It requires transition from one discrete energy level to another energy level.

$$E = E_2 - E_1 = hf = \frac{hc}{\lambda} \quad \text{where } h = 6.626 \times 10^{-34} \text{ J}$$

* when photon with energy $(E_2 - E_1)$ is incident on the material, an electron may be excited from E_1 into higher level E_2 through absorption of photon.

* Through emission of photon electron goes from higher level E_2 to lower level E_1 and energy loss.

*Einstein relation:

Boltzmann statistic: $\frac{N_1}{N_2} = \frac{g_1 \exp(-\frac{E_1}{KT})}{g_2 \exp(-\frac{E_2}{KT})}$

N_1 = density of atoms in energy level E_1 and N_2 = density of atoms in energy level E_2

$$\frac{N_1}{N_2} = \frac{g_1}{g_2} \exp\left(\frac{E_2 - E_1}{KT}\right)$$

g_1, g_2 = corresponding decay of level

$$\frac{g_1}{g_2} \exp\left(\frac{hf}{KT}\right)$$

$K = 1.381 \times 10^{-23} \text{ J/K}$

when decay of two level are equal

$g_1 = g_2$ then "

$\frac{\text{stimulated emission rate}}{\text{spontaneous emission rate}} = \exp\left(\frac{hf}{KT}\right)$

$\exp\left(\frac{hf}{KT}\right) - 1$

$1 - \exp\left(\frac{hf}{KT}\right)$

Math

Speed of light, $c = 2.998 \times 10^8 \text{ m/s}$
 Planck constant, $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
 Boltzmann's constant, $k = 1.381 \times 10^{-23} \text{ J/K}$

* Calculate the ratio of stimulated emission rate to the spontaneous emission rate for an incandescent lamp operating at a temperature of 1000 K . It may be assumed that the average operating wavelength is $0.5 \text{ }\mu\text{m}$.

Solution:

average operating frequency

$$c = f \lambda = \frac{2.998 \times 10^8}{0.5 \times 10^{-6}}$$

$$f = 6.0 \times 10^{14} \text{ Hz}$$

The ratio is

stimulated emission rate

spontaneous

$$\exp\left(\frac{hf}{kT}\right) - 1$$

$$= \frac{6.626 \times 10^{-34} \times 6 \times 10^{14}}{1000 \times 1.381 \times 10^{-23}} - 1$$

Thyridaceae

height = 13.1×10^{-13} not for

the body of the

transmission

length of a

maximum

to optical

to size, relative

to the operating